

MORADABAD INSTITUTE OF TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SESSIONAL TEST -1st (SET-1)

Course: B.Tech.

Sem: 2nd

Session: 2024-25

Time: 1 hr.

Sec: A, B, C and D

Subject: Programming For Problem Solving

Subject Code: BCS-201

Max. Marks: 20

Q.No. :	1	2	3	4	5	6
CO No. :	<u>1</u>	<u>1</u>	<u>2</u>	<u>2</u>	<u>2</u>	<u>3</u>
Bloom's Level	<u>K2</u>	<u>K2</u>	<u>K2</u>	<u>K3</u>	<u>K3</u>	<u>K3</u>

CO1: To develop simple algorithms for arithmetic and logical problems.

CO2: To translate the algorithms to programs & execution (in C language).

CO3: To implement conditional branching, iteration and recursion.

Note: All sections are compulsory.

SECTION- A

1. Explain the various primitive data types on the bases of range, format specifier and memory size used in C language. 2 marks

2. Explain the digital computer and its components with proper architecture (Block diagram). 3 marks

3. Explain bitwise and ternary operators in detail. What is the importance of precedence order and associativity. 3 marks

SECTION- B

4. WAP in C to check the entered year is leap year or not. Also write the syntax of if-else statement. 4 marks

5. WAP in C to perform arithmetic calculations (+, -, *, /) using switch statements. 4 marks

6. WAP in C to check whether the entered positive integer number is prime or not. 4 marks

9>b?b>c:a:b
ll:

if(n%i==0){
 cout<<endl;
}

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SESSIONAL TEST -1st (SET-1)

Time: 1 hr.

Course: B.Tech.

Sem: 1st

Session: 2025-26

Sec: E, F, G and H

Subject: Programming For Problem Solving

Subject Code: BCS-101

Max. Marks: 20

Q.No. :	1	2	3	4	5	6
CO No. :	1	1	1	2	2	2
Bloom's Level	K2	K3	K3	K1	K3	K3

CO1 To develop simple algorithms for arithmetic and logical problems.

CO2 To translate the algorithms to programs & execution (in C language).

CO3 To implement conditional branching, iteration and recursion.

Note:

1. This paper contains two sections (A) and (B)

2. All sections are compulsory.

SECTION- A

1.	Explain the bitwise and conditional operators in details.	2 marks
2.	Why break statement is used required? Write a C program to check the character is vowel or consonant.	3 marks
3.	Write a C program to check the entered year is leap year or not.	3 marks

SECTION- B

4.	Define the following in details: (i) Compiler (ii) Interpreter (iii) Linker (iv) Loader	4 marks
5.	Draw the architecture of Digital Computer System and its components in details.	4 marks
6.	Write a Program to check whether the entered number is prime number or not.	4 marks